

⚠ Problem with Long-CoT

- Visual Grounding Degradation
- Low Information Density Reasoning
- Token-Efficiency Collapse

🌀 DRT Representation (Ours)

- Separate perception from deduction
- Compress reasoning into dense telegraphic traces

<visual>	grounded observations
<think>	[Priors] -> dense telegraphic reasoning
<answer>	final answer

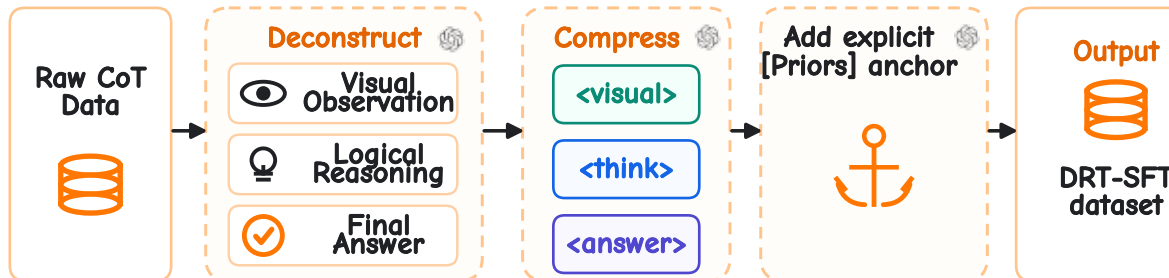
🎯 Outcome

- ✓ Faithful Visual Grounding
- ✓ Information-Dense Traces
- ✓ Token & Latency Reduction

Stage 1 SFT: Build DRT-format supervision

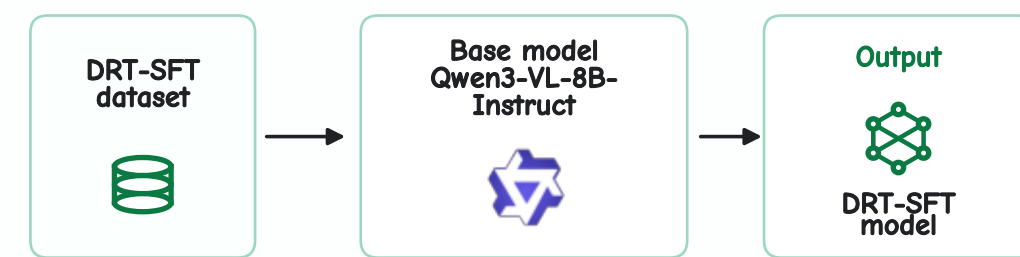
Transform raw CoT into DRT supervision for SFT initialization.

1 DRT-SFT Data Construction ★



2 SFT Initialization

Use DRT-style data for SFT to internalize dense reasoning.

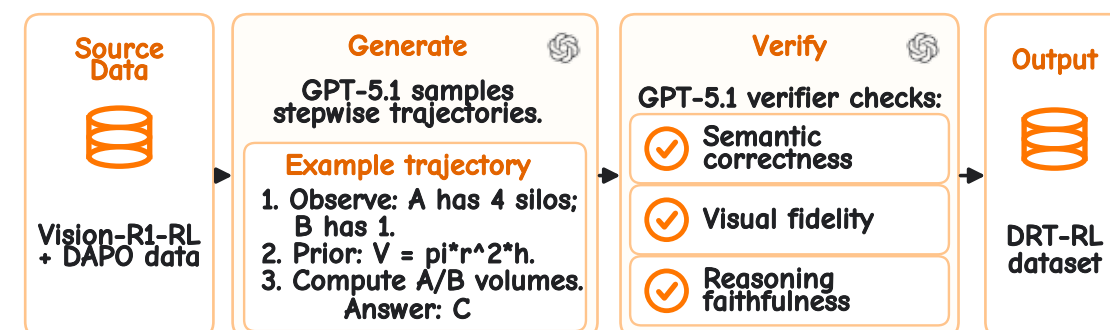


Stage 2 RL: Improve reasoning reliability under compact traces

Verified trajectories + process rewards improve compact reasoning.

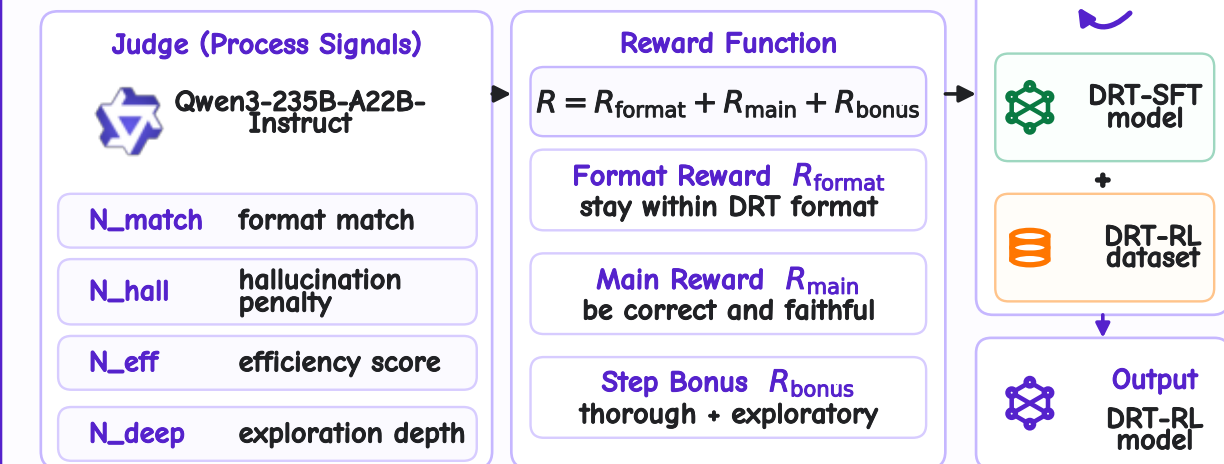
3 DRT-RL Data Construction ★

Generate & verify process-correct trajectories for RL training.



4 Reward Design + Process RL ★

Verified trajectories + process-level rewards improve reasoning.



5 DRT-trained Inference

